GATE ME 90 Days Challenge

Day 77/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
All Subjects	Mixed PYQ Day	22 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed set modelled on the most-repeated GATE ME previous-year concepts across all subjects: analogies and clock angles, Laplace shifting and Simpson's rule, the ballistic pendulum, thin-cylinder volumetric strain, Hooke's joint, transmissibility, belt power, jet impact on a moving plate, Prandtl number, entropy of the universe, the optimum Brayton pressure ratio, top-gate filling time and Little's law. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the pair that has the same relationship as AUTHOR : NOVEL

(A) Reader : Library
(B) Sculptor : Statue
(C) Teacher : Classroom
(D) Pen : Paper

Q2 NAT

The angle (in degrees) between the hour hand and the minute hand of a clock at 3:40 is _____

Q3 MCQ

The Laplace transform of $f(t) = t\,e^{-2t}$ is

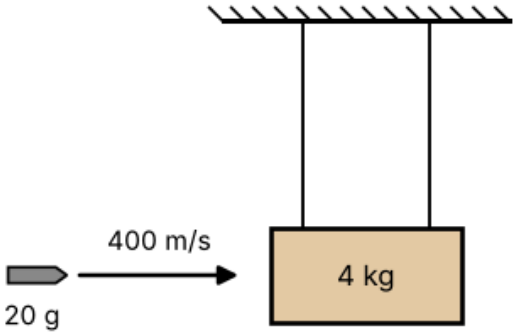
(A) $1/(s+2)^2$
(B) $1/(s-2)^2$
(C) $2/(s+2)^3$
(D) $s/(s+2)^2$

Q4 NAT

The integral of x^4 from $x = 0$ to $x = 2$ is evaluated by Simpson's 1/3 rule with a step size of 0.5. The value obtained (round off to three decimal places) is _____

Q5 NAT

A bullet of mass 20 g moving horizontally at 400 m/s embeds itself in a 4 kg wooden block hanging from light strings (ballistic pendulum), as shown. Taking g = 9.81 m/s², the height (in mm) through which the block rises (round off to one decimal place) is _____



Q6 MCQ

A thin cylindrical shell of diameter d and wall thickness t (closed ends) is subjected to an internal pressure p. With E the Young's modulus and ν the Poisson's ratio, its volumetric strain is

(A) (pd/(2tE))(1 − ν)
(B) (pd/(4tE))(1 − 2ν)
(C) (pd/(4tE))(5 − 4ν)
(D) (pd/(2tE))(5 − 4ν)

Q7 MCQ

The driving shaft of a Hooke's (universal) joint runs at a uniform 600 rpm, and the angle between the two shafts is 20°. The maximum and minimum speeds of the driven shaft are

(A) 600 rpm and 563.8 rpm
(B) 638.5 rpm and 600 rpm
(C) 621.2 rpm and 579.6 rpm
(D) 638.5 rpm and 563.8 rpm

Q8 MCQ

For a spring-mass-damper system excited by a harmonic force, the transmissibility equals 1 at a frequency ratio ω/ω_n of

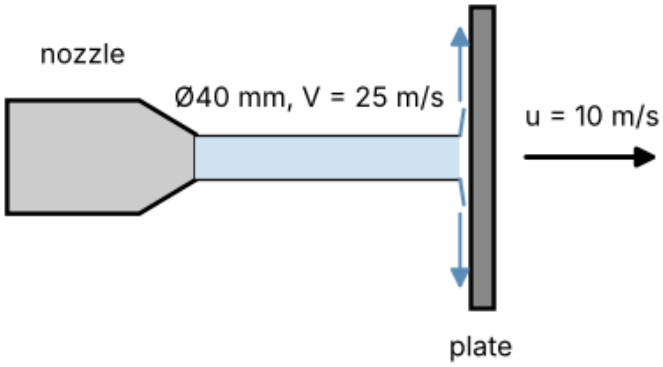
(A) 1, for every damping ratio
(B) √2, for every damping ratio
(C) √2 only when the system is undamped
(D) 2, for every damping ratio

Q9 NAT

An open flat-belt drive has an angle of contact of 165° on the smaller pulley and a coefficient of friction of 0.3. The belt speed is 15 m/s and the tension on the tight side is 2000 N (centrifugal tension neglected). The power transmitted (in kW, round off to two decimal places) is _____

Q10 NAT

A water jet of 40 mm diameter with a velocity of 25 m/s strikes normally a flat plate that moves away from the nozzle in the direction of the jet at 10 m/s, as shown. The force exerted by the jet on the plate (in N, round off to one decimal place) is _____



Q11 MCQ

For laminar flow over a flat plate, the hydrodynamic boundary layer and the thermal boundary layer have the same thickness when

(A) the Prandtl number is 1
(B) the Reynolds number is 1
(C) the Nusselt number is 1
(D) the Biot number is less than 0.1

Q12 NAT

1 kg of water (c = 4.18 kJ/kg·K) at 20 °C is placed in contact with a large reservoir at 80 °C until it reaches 80 °C. The entropy change of the universe (in kJ/K, round off to three decimal places) is _____

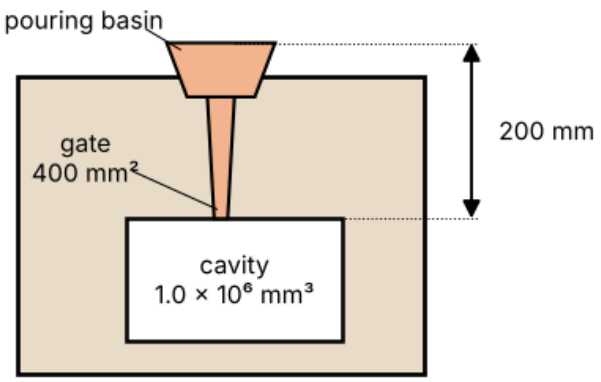
Q13 MCQ

An ideal Brayton cycle with air (γ = 1.4) works between a minimum temperature of 300 K and a maximum temperature of 1200 K. The pressure ratio that gives the maximum net work per kg of air is about

(A) 4.0
(B) 8.0
(C) 11.3
(D) 16.0

Q14 NAT

A mould cavity of volume 1.0×10^6 mm³ is filled through a top gate, as shown. The height of molten metal above the gate (sprue height) is 200 mm and the gate area is 400 mm². Neglecting friction and taking g = 9.81 m/s², the filling time (in s, round off to two decimal places) is _____



Q15 MCQ

A machine shop holds on average 30 jobs as work-in-process, and it completes 6 jobs per day in steady state. The average time a job spends in the shop is

(A) 0.2 days
(B) 180 days
(C) 36 days
(D) 5 days

Answer Key

Q	Type	Answer
Q1	MCQ	(B) Sculptor : Statue
Q2	NAT	130
Q3	MCQ	(A) $1/(s + 2)^2$
Q4	NAT	6.417 (accept 6.38 to 6.45)
Q5	NAT	201.9 (accept 200.8 to 202.9)
Q6	MCQ	(C) $(pd/(4tE))(5 - 4\nu)$
Q7	MCQ	(D) 638.5 rpm and 563.8 rpm
Q8	MCQ	(B) $\sqrt{2}$, for every damping ratio
Q9	NAT	17.36 (accept 17.20 to 17.50)
Q10	NAT	282.7 (accept 281.0 to 284.5)
Q11	MCQ	(A) the Prandtl number is 1
Q12	NAT	0.068 (accept 0.067 to 0.070)
Q13	MCQ	(C) 11.3
Q14	NAT	1.26 (accept 1.25 to 1.27)
Q15	MCQ	(D) 5 days

Detailed Solutions

Q1 **Answer: (B) Sculptor : Statue**

Concept: In analogy questions, state the relation in one sentence and test each option with the same sentence.

'An author creates a novel.' Only 'a sculptor creates a statue' fits the same sentence.

A reader uses a library, a teacher works in a classroom, and a pen writes on paper: none is creator → creation.

Answer: **(B)**

Exam tip: If two options seem to fit, make the relation sentence more specific (creator–creation, part–whole, worker–tool).

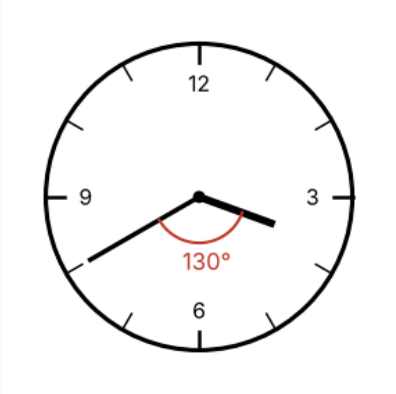
Q2 **Answer: 130**

Concept: The minute hand moves 6° per minute and the hour hand 0.5° per minute, so $\theta = |30H - 5.5M|$.

$\theta = |30 \times 3 - 5.5 \times 40| = |90 - 220| =$ **130°**

Check: minute hand at 240°, hour hand at 90 + 20 = 110°, difference 130° (clock face below).

Exam tip: If the formula gives more than 180°, subtract it from 360° to get the smaller angle.



Q3 **Answer: (A) $1/(s+2)^2$**

Concept: First shifting theorem: $L\{e^{at} f(t)\} = F(s - a)$.

$L\{t\} = 1/s^2$; with $a = -2$ the shift gives $F(s + 2) = 1/(s + 2)^2$.

Answer: **(A)**

Exam tip: $L\{t^n e^{at}\} = n!/(s - a)^{n+1}$; watch the sign of a when shifting.

Q4 **Answer: 6.417 (accept 6.38 to 6.45)**

Concept: Simpson's 1/3 rule: $I \approx (h/3)[y_0 + 4(y_1 + y_3) + 2y_2 + y_4]$, with an even number of strips.

$x = 0, 0.5, 1, 1.5, 2 \rightarrow y = 0, 0.0625, 1, 5.0625, 16$

$I \approx (0.5/3)[0 + 4(0.0625 + 5.0625) + 2 \times 1 + 16] = (1/6)(20.5 + 2 + 16) = 38.5/6 =$ **6.417**

The exact value is 32/5 = 6.4; the error comes from the fourth derivative (24), which is not zero.

Exam tip: Simpson's 1/3 rule is exact for polynomials up to degree 3, not degree 4.

Q5 **Answer: 201.9 (accept 200.8 to 202.9)**

Concept: The impact is perfectly plastic: momentum is conserved during impact (energy is not); after impact, energy is conserved during the swing.

Common velocity $V = mv/(m + M) = 0.02 \times 400/4.02 = 1.990$ m/s

$h = V^2/(2g) = 1.990^2/(2 \times 9.81) = 0.2019$ m = **201.9 mm**

Exam tip: Never equate the bullet's kinetic energy to (m + M)gh: more than 99 % of it is lost in the plastic impact.

Q6 **Answer: (C) $(pd/(4tE))(5 - 4\nu)$**

Concept: $\sigma_h = pd/(2t)$, $\sigma_l = pd/(4t)$; volumetric strain $= 2\varepsilon_h + \varepsilon_l$.

$\varepsilon_h = (pd/(4tE))(2 - \nu)$, $\varepsilon_l = (pd/(4tE))(1 - 2\nu)$

$\varepsilon_v = (pd/(4tE))(4 - 2\nu + 1 - 2\nu) =$ **$(pd/(4tE))(5 - 4\nu)$**, answer **(C)**

Exam tip: For a thin sphere, $\varepsilon_v = 3\varepsilon = (3pd/(4tE))(1 - \nu)$.

Q7 **Answer: (D) 638.5 rpm and 563.8 rpm**

Concept: For a Hooke's joint, $\omega_{max} = \omega/\cos\alpha$ and $\omega_{min} = \omega\cos\alpha$.

$N_{max} = 600/\cos 20° = 600/0.9397 = 638.5$ rpm

$N_{min} = 600 \times 0.9397 = 563.8$ rpm. Answer **(D)**

Exam tip: The maximum speed occurs twice per revolution; a double Hooke's joint (equal angles, forks in one plane) removes the fluctuation.

Q8 **Answer: (B) $\sqrt{2}$, for every damping ratio**

Concept: $TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$.

$TR = 1$ requires $(1 - r^2)^2 = 1$, i.e. $r^2 = 2$, $r = \sqrt{2}$, independent of ζ: all the TR curves cross at $r = \sqrt{2}$.

Answer: **(B)**

Exam tip: Isolation ($TR < 1$) needs $r > \sqrt{2}$, and there more damping slightly increases TR.

Q9 **Answer: 17.36 (accept 17.20 to 17.50)**

Concept: $T_1/T_2 = e^{\mu\theta}$ with θ in radians; $P = (T_1 - T_2)v$.

$\theta = 165 \times \pi/180 = 2.880$ rad; $e^{0.3 \times 2.880} = e^{0.864} = 2.372$

$T_2 = 2000/2.372 = 843.0$ N

$P = (2000 - 843.0) \times 15 = 17\,355$ W = **17.36 kW**

Exam tip: Use the angle of contact on the smaller pulley, where slip occurs first.

Q10 **Answer: 282.7 (accept 281.0 to 284.5)**

Concept: For a single moving plate, the mass striking per second is $\rho A(V - u)$ and the relative velocity destroyed is $(V - u)$, so $F = \rho A(V - u)^2$.

$A = \pi/4 \times 0.04^2 = 1.2566 \times 10^{-3}$ m²

$F = 1000 \times 1.2566 \times 10^{-3} \times (25 - 10)^2$ = **282.7 N**

The work done on the plate is $F \times u = 2827$ W.

Exam tip: For a series of plates (a wheel), the whole jet ρAV strikes, so $F = \rho AV(V - u)$ instead.

Q11 **Answer: (A) the Prandtl number is 1**

Concept: $\delta_t/\delta \approx Pr^{-1/3}$, because $Pr = \nu/\alpha$ compares momentum diffusion with heat diffusion.

When $Pr = 1$, momentum and heat diffuse at the same rate, so $\delta_t = \delta$.

Answer: **(A)**

Exam tip: Liquid metals ($Pr \ll 1$) have $\delta_t \gg \delta$; oils ($Pr \gg 1$) have $\delta_t \ll \delta$.

Q12 **Answer: 0.068 (accept 0.067 to 0.070)**

Concept: $\Delta S_{universe} = \Delta S_{water} + \Delta S_{reservoir}$; the reservoir stays at constant temperature.

$\Delta S_{water} = mc\ln(T_2/T_1) = 4.18\ln(353.15/293.15) = 0.7784$ kJ/K

$\Delta S_{reservoir} = -Q/T = -4.18 \times 60/353.15 = -0.7102$ kJ/K

$\Delta S_{universe}$ = **0.068 kJ/K** (positive, as heat flows through a finite temperature difference)

Exam tip: Always use kelvin in $\ln(T_2/T_1)$ and in Q/T.

Q13 **Answer: (C) 11.3**

Concept: Net work is maximum when $T_2 = T_4 = \sqrt{(T_{min} T_{max})}$, giving $r_p = (T_{max}/T_{min})^{\gamma/(2(\gamma - 1))}$.

$r_p = 4^{1.4/0.8} = 4^{1.75} =$ **11.3**, answer **(C)**

At this ratio, $T_2 = T_4 = 600$ K.

Exam tip: At the pressure ratio $(T_{max}/T_{min})^{\gamma/(\gamma - 1)}$ (= 128 here) the net work falls to zero.

Q14 **Answer: 1.26 (accept 1.25 to 1.27)**

Concept: In a top gate the head stays constant during filling (metal falls freely into the cavity), so $v = \sqrt{(2gh)}$ throughout and $t = V/(A\,v)$.

$v = \sqrt{(2 \times 9.81 \times 0.2)} = 1.981$ m/s

$t = 1.0 \times 10^{-3}\ m^3/(400 \times 10^{-6}\ m^2 \times 1.981\ m/s) =$ **1.26 s**

Exam tip: For a bottom gate the effective head falls as the cavity fills, and $t = (A_m/A_g)(2/\sqrt{(2g)})(\sqrt{h_t} - \sqrt{(h_t - h_m)})$, which is longer.

Q15 **Answer: (D) 5 days**

Concept: Little's law: $L = \lambda W$, valid for any stable system regardless of distributions.

$W = L/\lambda = 30/6 =$ **5 days**, answer **(D)**

Exam tip: Little's law also gives $L_q = \lambda W_q$; it is the quickest check on any queuing answer.

What next?

Day 77 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 78/90: Full-Length Mock 2. Comment **GATE90** on our reel to get all links.
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